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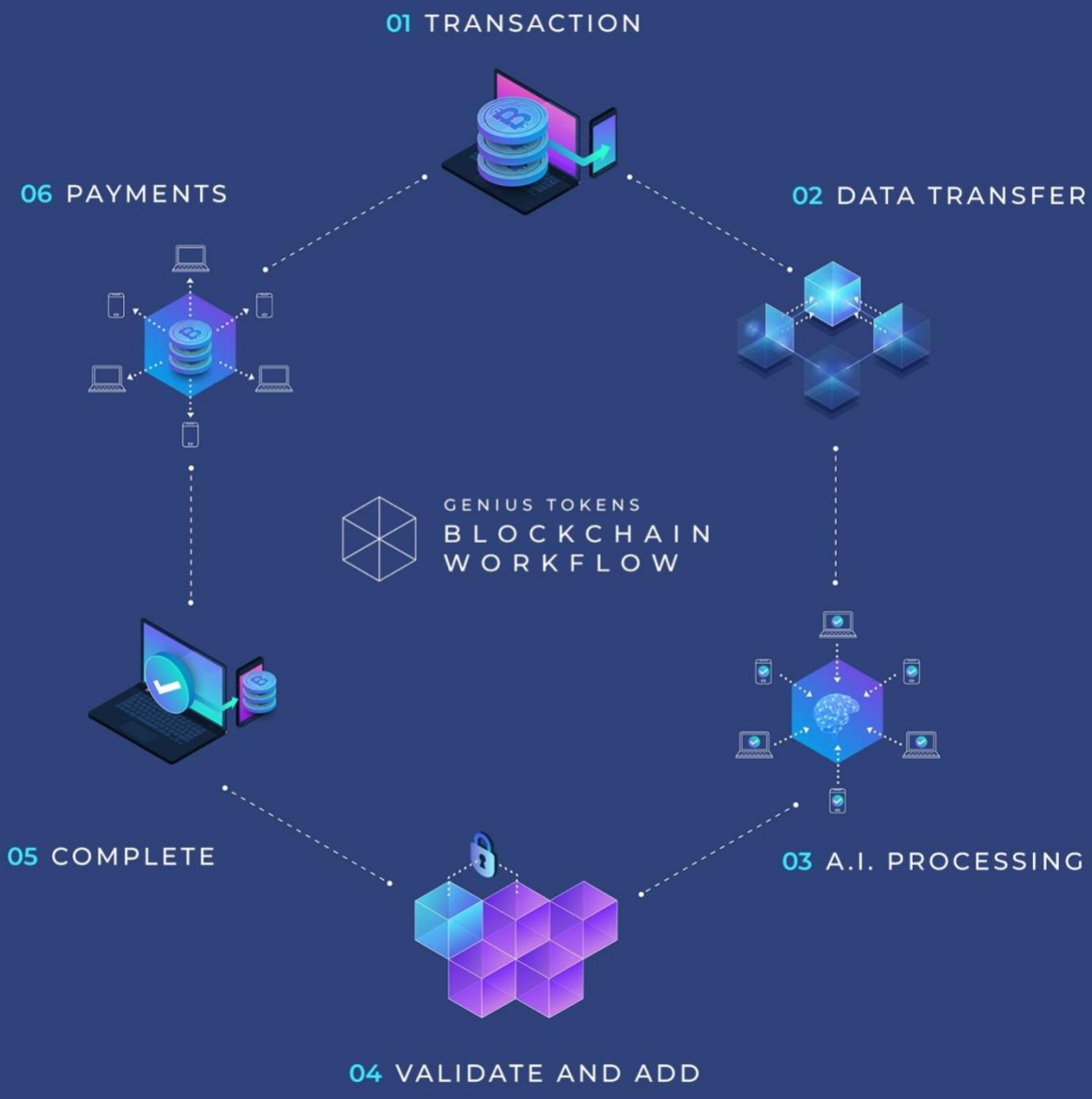


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Abstract

Historically, access to high-performance computer systems was reserved for large companies that have the technical, financial, and human resources to own and manage these systems. Over 5 billion consumers interact with data every day, and Samsung estimates that this will be 6 billion by 2025. Developers need better tools and methods to handle the additional processing required for high-volume data processing, which has intensified the need for advanced computing and performance.

In response to this growing demand, companies such as Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP) recently introduced cloud-based computing services for small and medium-sized developers who need advanced IT resources to support a wide range of applications. Although cloud computing resources can support global applications, the centralized architecture introduces the issue of scalability.

Distributed computing solves the issue of scalability by spreading the demand of compute processes across several high-powered machines.

The Genius tokens system infrastructure serves as a distributed multi-purpose service with a cryptographic payment system that allocates computing resources and integrates them into real-world projects. By purchasing Genius Tokens, developers buy as many scalable distributed resources as necessary to power their projects. These tokens can be used in exchange for computing power, or they can be exchanged for conventional currency.

By tokenizing computing power, developers can integrate their projects on blockchain technology and leverage its inherent distributed architecture. Genius Token integration provides:

- A hybrid cryptocurrency and crypto token system allowing for high-speed transactions decoupled from slower cryptocurrency architecture.
- Similar to cryptocurrency in-game purchases, payments and data resources are tied to the end-user’s wallet.
- Transfer of data directly to the client or customer.

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Background

The growth of data processing and access requires improvements in infrastructure design. Although changes can be gradual, they must support better performance as technology evolves in every industry and more users expect high performance from any application regardless of its growth.

The International Data Corporation (IDC) predicts that the high-performance computing (HPC) sector will experience a compound annual growth rate (CAGR) of 8%, which raised the total market to $31.4 billion in 2019. Rendering and visualization software grew at a CAGR of 30.03% from 2016 to 2020 with 3D visualization devices, virtual and augmented reality, and high-end video games fueling that growth. Big data and business analysis revenue worldwide reached $150.8 billion in 2017, an increase of 12.4% compared to 2016. The commercial purchase of BDA hardware, software and services maintained a CAGR of 11.9% in 2020, when the turnover exceeded $210 billion dollars.

For example, the market for neural network software is expected to increase from $7.17 billion in 2016 to $ 22.55 billion by 2021. This is due to unclassified and recently digitized data since neural networks, with intensive use of computers, are classifying machines.

Advanced risk analysis has a CAGR of 15.3%, which increases the market from $17.60 billion in 2017 to $35.92 billion in 2022. New technologies, such as artificial intelligence, allow institutions to improve underwriting decisions and increase revenue while reducing risk costs. Computer and Block Desktop grids can potentially interrupt all clouds, large data, and HPC economics. The IDC also forecasts worldwide spending on cognitive and artificial intelligence systems will reach $57.6 Billion in 2021.

Currently, several solutions for high-speed processing of transactions via a blockchain crypto token are available. Some of these solutions attempt to use different techniques to speed up the verification of transactions, but these solutions fail to meet the needs of the industry since current systems take up to minutes for transaction verification. Other solutions attempt to solve this issue by using direct acyclic graph (DAG) blockchain crypto tokens, but still fall short as decentralized applications are not integrated.

Distributed General-Purpose Computing with a Crypto Token Payment System

Genius Ventures aims at providing a system and method for distributed general-purpose computing with a crypto token payment system. This payment system integrates a slow blockchain crypto token with a fast DAG-based crypto token. This technology is based on smart contracts "Genius Tokens" from Ethereum and allows the creation of a virtual cloud infrastructure that provides high-performance computing services on demand.

Genius Ventures seeks to implement a scalable, high-performance, secure, and manageable infrastructure side chain that promotes a new form of distributed management, which includes key leaders in computing, large data, and cloud industries.

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We believe in the future of decentralized infrastructures and market networks. A future where big data and high-performance computing (HPC) applications, high-value data sets, and computing resources are monetized on blockchain with the highest level of transparency, resistance, and security.

Genius Venture will play a critical role in this future by building a highly scalable, hybrid blockchain solution.

Market Challenges and Opportunities

Current Limitations

Blockchains like Ethereum offer a new approach to the work of decentralized applications (DApps). Ethereum allows developers to write smart contracts and code applications that run on a virtual machine (VM). This represents a potential revolution in the design and implementation of services such as investments, finance, corporate financing, internet, insurance, forecasting markets, gambling, distributed data processing and more.

The difference between traditional applications running on a VM and blockchain applications is that blockchain requires minimal computing resources. Blockchain applications only need a few kilobytes of storage, a low-resource VM and a high-latency protocol.

Existing clouds cannot meet the DApp requirements of decentralized infrastructure. At the same time, industry and scientific communities are increasing their demand for more computing power to deploy large-scale applications and manage large amounts of data.

Applications that manage big data and use artificial intelligence, machine learning, and other resource intensive applications need high performance computer power, but small innovative development teams often lack the funds and experience to provision, support, and use them. Even though cloud providers reduce costs compared to hosting on-premises resources, the cost to provision high-powered resources is still high.

Currently, data centers spend extensive amounts of energy running servers and cooling systems. Not only is it expensive, but it can also have a significant negative impact on the environment. We need a new form of decentralized cloud that can use blockchain computing and reduce the cost of using the infrastructure.

The Genius Venture Solution

Technical Overview

Genius Venture supports the emerging class of blockchain-based distributed applications and enables cost-effective, high-performance computing by building decentralized cloud infrastructure. The infrastructure uses DAG-based blockchain crypto tokens integrated with decentralized applications.

Our system is unique when compared to other known systems and solutions in that it provides a full system that integrates a slow blockchain crypto token with a fast DAG-based blockchain crypto token. The software is unique when compared with other known solutions by providing a way for applications and games to utilize unused cycles from computing devices. Developers generate crypto tokens by processing data using artificial intelligence or machine learning on devices. Typically, other cryptocurrencies and crypto tokens are earned by what is known as mining. Mining uses computing devices to solve complex mathematical problems to verify a block on the blockchain. This mining does not process real data and is only used competitively to identify who completes the mathematical calculation first to determine who wins mining payments.

A blockchain-based decentralized cloud connected with a hybrid cryptocurrency and crypto token payment system allows on-demand, secure, and low-cost access to the most competitive computing infrastructures. DApps rely on Genius Ventures to automatically search, find, provision, use, and release all computing resources necessary for applications and their data.

We envision a new ecosystem of companies offering storage, server farms, data providers, web hosting, and SaaS applications, all conducting business with each other through Genius Ventures. The distributed cloud can open new markets for aggressive use of existing computing infrastructures.

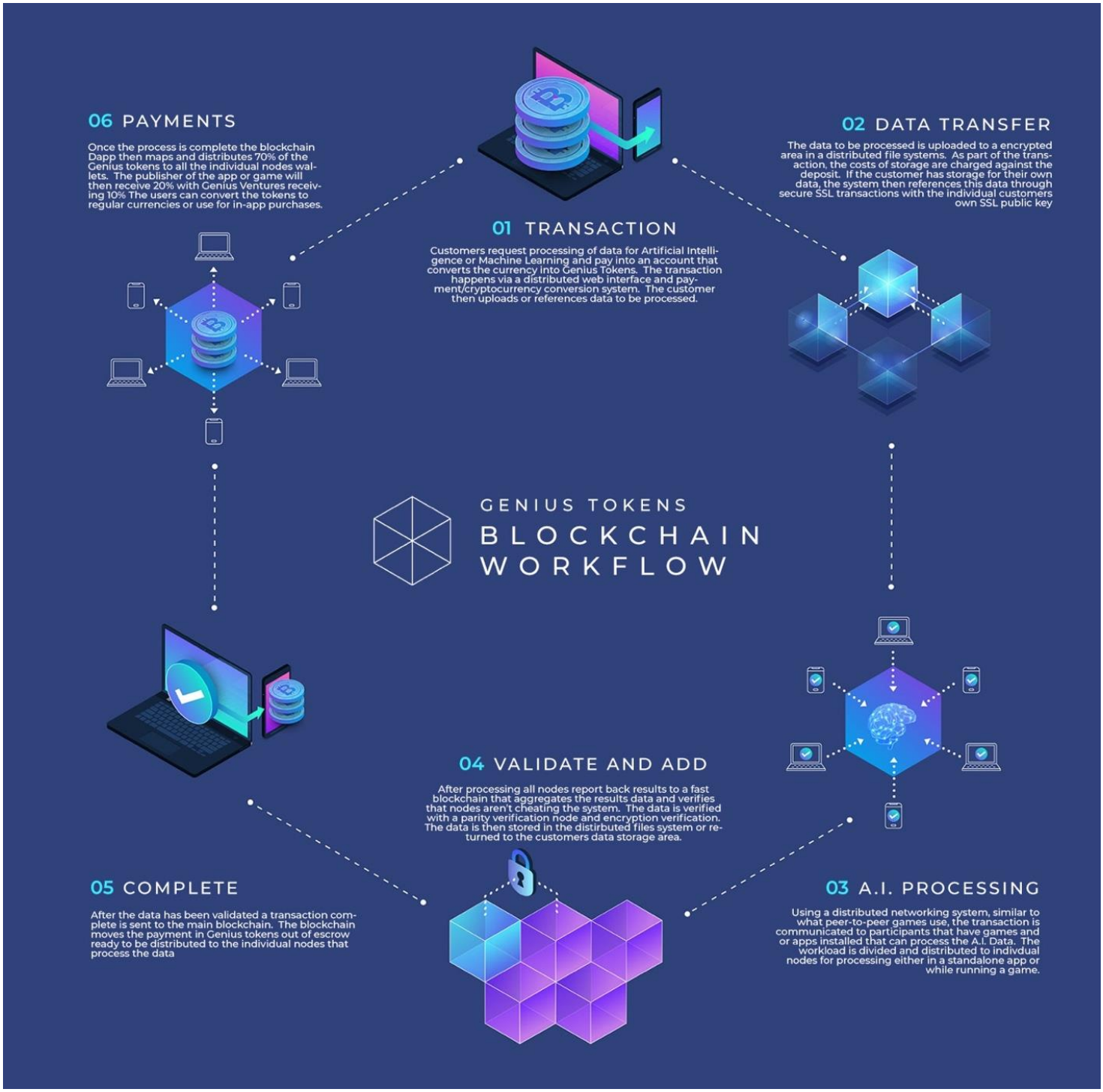
Core Value Proposition

At Genius Ventures, we aim to address the needs of all decentralized businesses outlined below.

1. DApp providers can perform off-chain computations on demand.
2. Application suppliers lower computing prices of their decentralized applications by employing a safe, reliable, and robust infrastructure.
3. The customer can select artificial intelligence (AI) or machine learning (ML) algorithms through a customer portal on the customer's secure servers.
4. Stand-alone applications, embedded systems or games with the SDK integrated can process instructions.
5. Server suppliers provide their servers, leverage underused computing resources, and increase their return on investment on existing infrastructure. The hybrid crypto token system uses a fast internal DAG-based blockchain that executes transactions in microseconds.
6. Code and SDK run on all devices, including Windows, OSX, iOS, Android, Linux, Windows Mobile, Xbox, PlayStation, Nintendo and IoT.
7. User setup costs are almost zero and users get paid to process data. Both incentivize users to use your game or app.

Genius Token Blockchain

Utilizing unused cycles on servers and mobile and IoT devices, the system processes AI and ML data on the target device. The end-user gets paid in Genius crypto tokens that can be reused for in-app purchases or converted back to other currencies. The following workflow demonstrates the way Genius Token blockchain executes:



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Market Overview - Opportunities

The Perfect Timing

The convergence of several trends has created an optimal business opportunity for a decentralized cloud infrastructure:

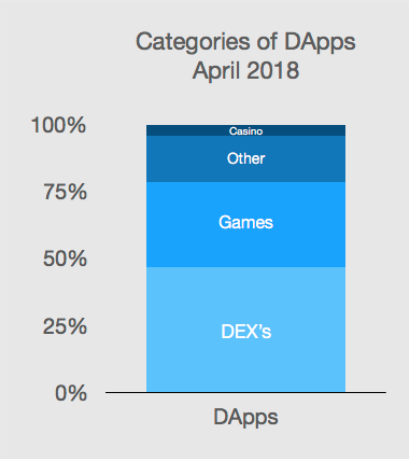
- An introduction of better DApp performance and blockchain integration.
- Leveraging unused resources for pay generates a better return on investment for server providers, leaving a large supply of potential resource partners.
- Smart Contracts simplify infrastructure of decentralized computing resources, making them the preferred way to build distributed applications.

The sum of these trends carves a niche in our target market to launch better cloud computing initiatives to prepare for the growing demand of high-performance cloud resources.

The DApps Market

In 2018, Chris McCann carefully analyzed, scrutinized, and categorized the largest decentralized application platform -- the Ethereum blockchain and its (at the time) 312 DApps. He categorized the DApps into four main categories:

- Games
- Pyramid schemes
- Decentralized exchanges
- Casino applications



Source: McCann (April 2018)

The keynote of his research focused on the fact that the top DApps in the categories listed above are minimal in comparison to the traditional web and existing mobile applications. He found developers were far from consumer adoption of DApps since no specific designated or trending applications were developed aside from tokens or trading. McCann further established the fact that any seemingly large DApp (e.g., IDEX) has low usage overall and concluded that, “as an ecosystem, we need to build better tools and infrastructure, if we need to attain global adoption of DApps.”

Genius Ventures positions itself as an engineering tool, by allowing DApps to have secure, scalable, and straightforward access to computing resources. We believe these resources and innovations ensure computational support for a wide superfluity of CPU or GPU-intensive DApps in the fields of

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artificial intelligence, gaming, cryptography, 3D rendering, and scientific computing. By focusing on DApps integrated into a hybrid cryptocurrency/crypto token system, we envision a crypto sphere of more valuable and diversified applications that make use of unique properties of blockchain and grow throughout their journey to compete with traditional consumer web applications.

The Edge and Fog Computing Market

Fog computing is a parallel, system-level architecture that distributes computing, storage, control, and networking functions closer to users along a cloud-to-thing continuum. According to research by OpenFog, the global fog computing market has the potential to reach more than $18 billion worldwide by 2022.

The Internet is rapidly increasing its potential to transform everyday life in homes, cities, farms and smart manufacturing facilities. The prospects for growth in the market are enormous, and Gartner estimates that 20.4 billion related articles were used worldwide by 2020.

The development of an Internet-based solution requires cooperation, coordination, and connectivity without compromising any part of the system. All devices must work together and integrate with all other devices and these devices must interact and communicate with connected systems and infrastructure transparently and securely. This is possible to achieve, but it can be costly, time-consuming, and difficult unless new thinking and IoT security approaches emerge from the current centralized model.

System Architecture Overview

The Genius Venture Invention

The Genius Venture invention is a system and method for distributed general-purpose computing that integrates a slow blockchain crypto token with a fast DAG-based blockchain crypto token. The innovative system provides:

- A hybrid cryptocurrency/crypto-token system that allows for high-speed transactions decoupled from the slow transactions of normal cryptocurrencies.
- A mechanism to verify the processing of data and payments to the end user’s wallet since the processing of that data can, in turn, be used for in-app or in-game purchases.
- It provides mechanisms for the transfer of unprocessed and processed data to the client or customer.

Similarly, the associated software is unique in that it incorporates interfaces that allow for applications and games to use unused cycles of one or more devices to process real data and to receive and utilize payments for the processing of data.

Our Component Process

The system is made up of the following components:

- Generic Blockchain Cryptotoken
- Directed Acyclic Graph Blockchain Cryptotoken
- Payment System
- Proof of Work
- Decentralized File System
- Data Delivery/Storage
- Processing
- Communications
- Software Development Kit
- Cryptocurrency Wallet
- Compute Devices

The associated computer process is made up of the following executable steps, all of which are required in all versions:

1. A customer deposits currency into a converter system that converts it into a crypto token and sends the transaction to the Generic Blockchain Cryptotoken component, which holds the payment as a deposit. The customer then sends the data to be processed to the Decentralized File System component which is stored in the Delivery/Storage component.
2. The customer then initiates a processing request with the Decentralized File System components' location and the Generic Blockchain Cryptotoken deposit account. The Generic Blockchain Cryptotoken component calculates out the cost of processing and puts that amount of crypto token on hold for processing.
3. The Generic Blockchain Cryptotoken component then deposits into the Directed Acyclic Graph Blockchain component equivalent amounts of the Directed Acyclic Graph Blockchain crypto token into the Direct Acyclic Blockchain.
4. The Directed Acyclic Graph Blockchain Cryptotoken component then sends the processing request to the Communication component that distributes the work to a multitude of end-user devices for processing through the Software Development Kit.
5. The Software Development Kit component then sends the request to the processing component.
6. The processing component then communicates with compute devices to process the data.
7. Compute devices process the data and send the processed data to the delivery/storage component then signals back to the processing component the process is complete.
8. The processing component then signals back to the Software Development Kit component on process completion.
9. The Software Development Kit component then signals back to the communication component process completion for one device.
10. Once process completion is achieved, the communication component sends a signal back to the Proof-of-Work component that verifies the integrity of the processed data and signals the Generic Blockchain Cryptotoken component to release the hold on the Generic Blockchain Cryptotoken.

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11. The Generic Cryptotoken then sends a signal to the Directed Acyclic Graph Blockchain token to release Cryptotokens to the end-user's crypto token wallet.

Distributed General-purpose Computing System with Cryptotoken Payment System

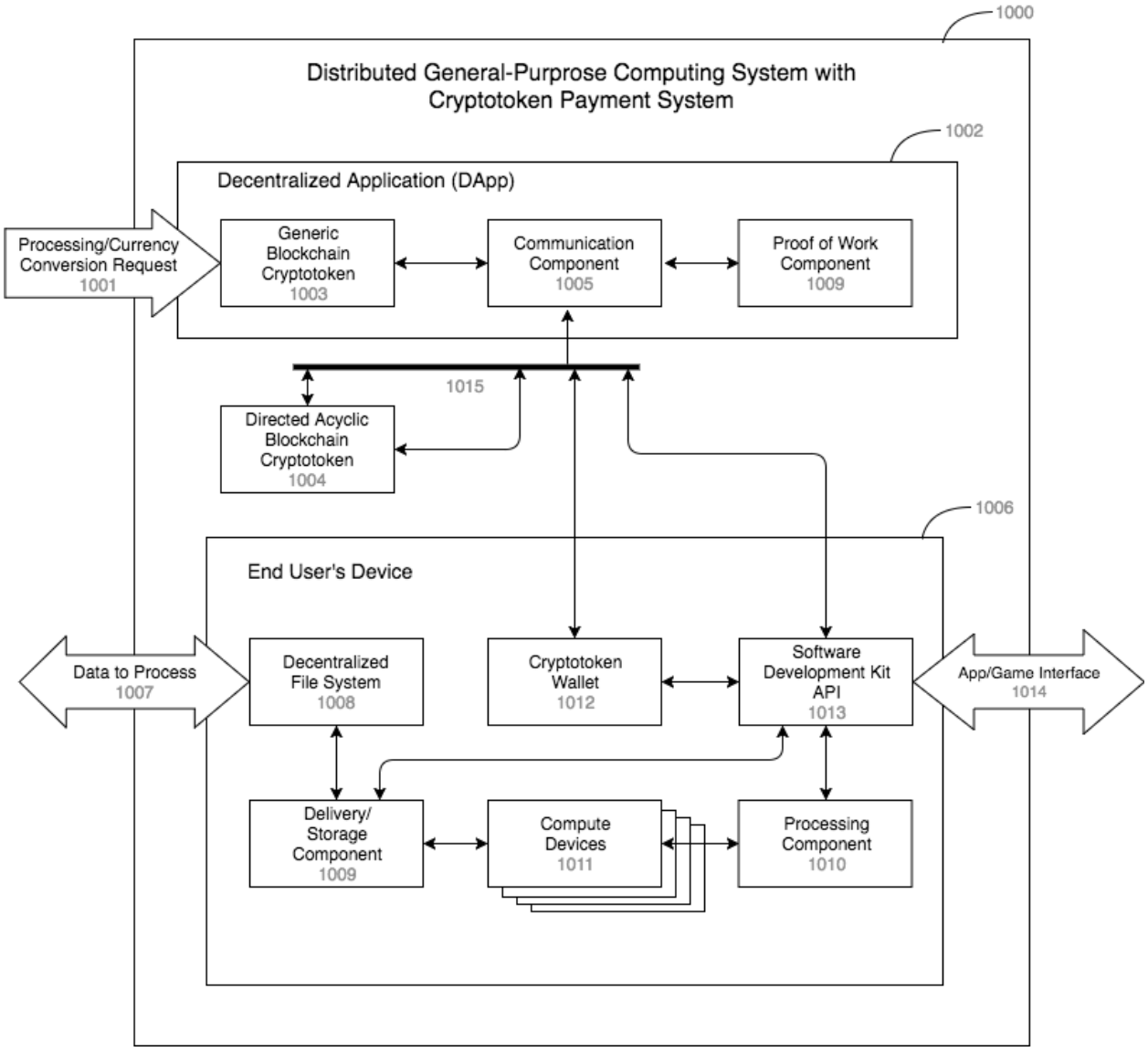


FIG. 1 – shown above is a schematic diagram that illustrates a Distributed General-Purpose Computing System with the Cryptotoken Payment System 1000, according to an embodiment.

In some embodiments, a Distributed General-Purpose Computing System with Cryptotoken Payment System is referred to as a cryptocurrency payment system for General Compute Devices or a Decentralized Crypto Payment system for Computing.

Decentralized Application

As shown in FIG. 1 above, the Distributed General-Purpose Computing System 1000 includes a Decentralized Application 1002, an End User's Device 1006 and a Directed Acyclic Graph Blockchain Cryptotoken Component 1004. The Decentralized Application 1002 is in communication with the End User's Device 1006 and the Directed Acyclic Graph Blockchain Cryptotoken 1004 through the Communication Component 1005 and via a network 1015. The network 1015 can be any network (e.g., a local area network (LAN), an extensive area network (WAN), a virtual network, a telecommunications network, a cellular network, a wireless LAN (WLAN), etc.) configured to enable communication between the Decentralized Application 1002, the Directed Acyclic Graph Blockchain Cryptotoken component 1004, and the End User's Device 1006. The network 1015 can be implemented as a wired network, a wireless network, or any other communication such as Bluetooth, NFC, and others.

The Decentralized Application 1002 contains three components, namely:

- Generic Blockchain Cryptotoken component 1003
- Communication Component 1005
- Proof of Work component 1009

The End User's Device 1006 has six components, namely:

- Decentralized File System 1008
- Cryptotoken Wallet 1012
- Software Development Kit API 1013
- Delivery/Storage component 1009
- Processing component 1010

The Directed Acyclic Graph Blockchain Cryptotoken 1004 is a stand-alone component.

Internal Components of the Software Development Kit API

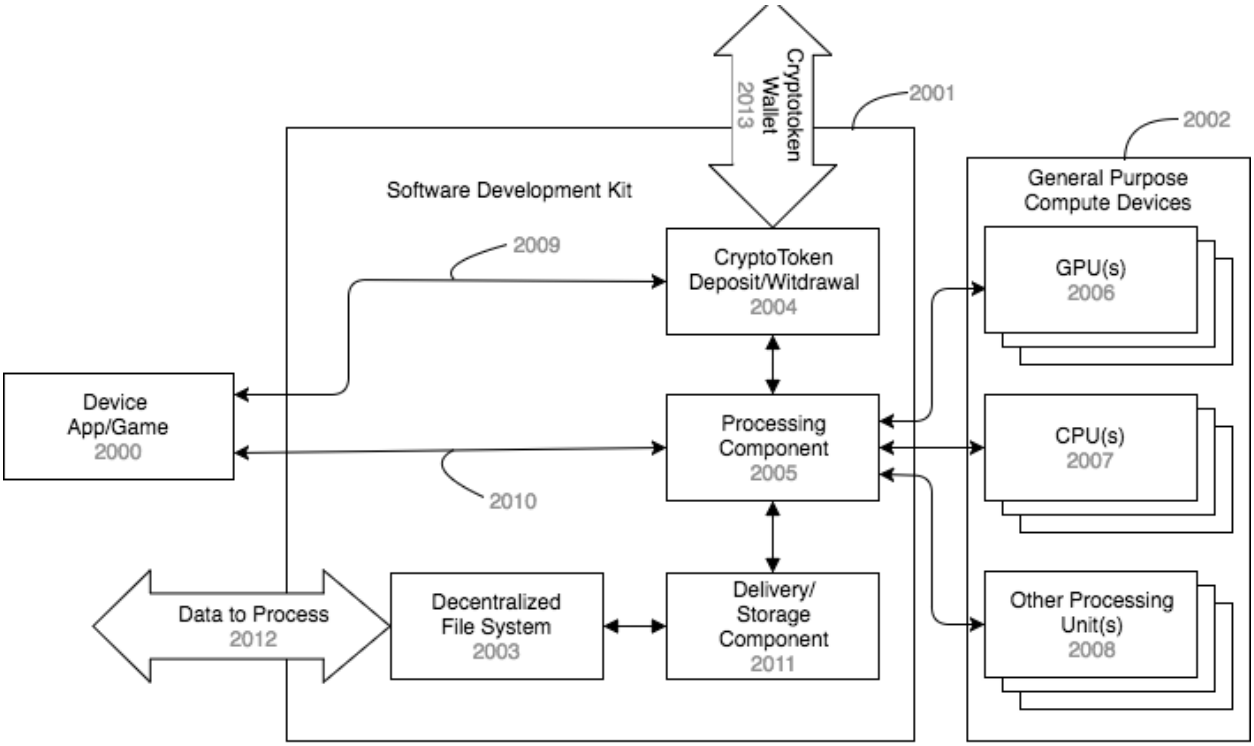


FIG. 2, as shown above, is a schematic diagram of the internal components of the Software Development Kit 2001 and the Software Development Kit API 1013. The General-Purpose Compute Device 2002 is a representation of any computing device. These can include Graphics Processing Units (GPU(s)) 2006, Central Processing Units (CPU(s)) 2007, and other types of Processing Units 2008. A Device App/Game 2000 connected to the Software Development Kit 2001 via an Application Programming Interface (API) 2009, 2010. For more details on how the internal components work, please refer to our technical paper.

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Customer Transaction

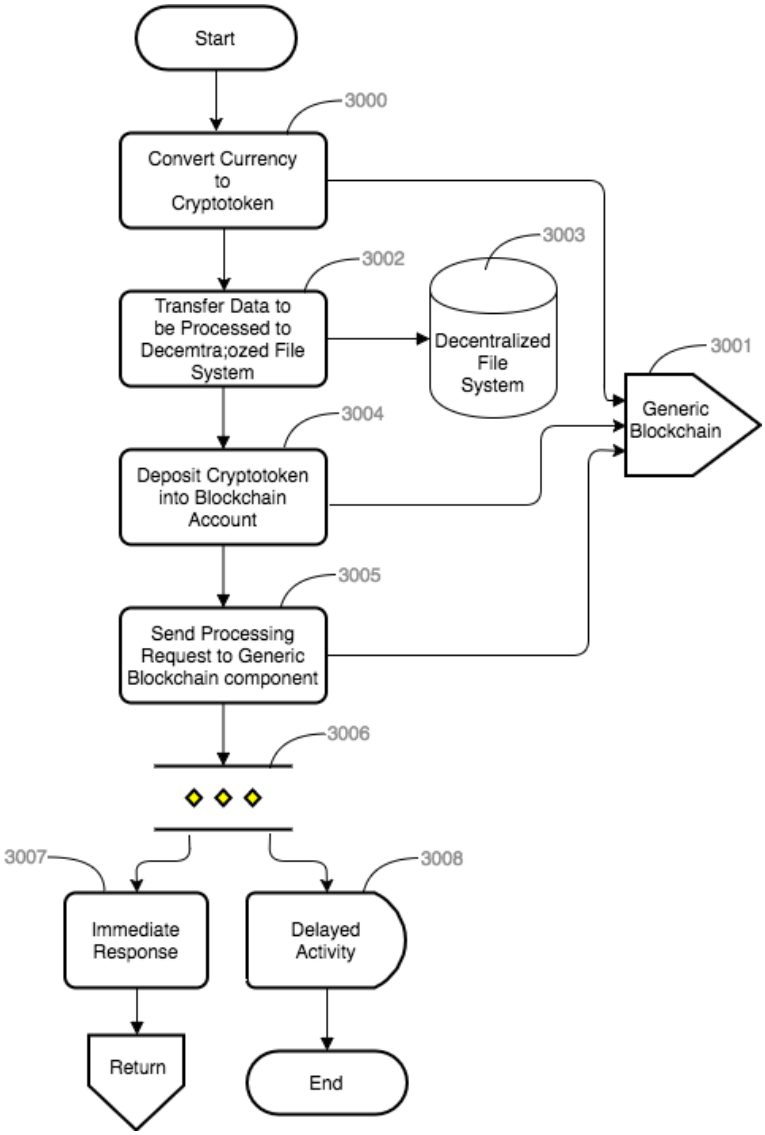


FIG. 3 is a flow diagram of a typical client or customer transaction. A customer or client converts any currency to a Cryptotoken 3000. This token is sent to the Generic Blockchain 3001.

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The Internal Workings of Request Processing

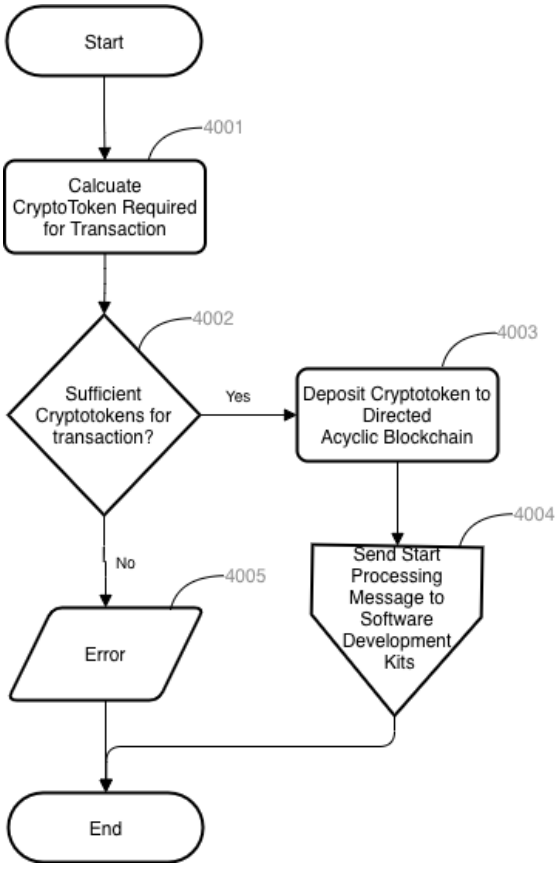


FIG. 4 is a flow diagram of the internal workings of the processing request inside the DApp. The DApp calculates the amount of crypto token required for this transaction 4001. The calculation uses the amount of processing power needed and the size of the data to be processed in determining if there is enough crypto token on deposit to execute this transaction 4002. If there are sufficient crypto tokens available, the Cryptotokens deposit into the Directed Acyclic Graph Blockchain 4003. The DApp then sends the processing request to the Software Development Kit API 4004. If there are not enough crypto tokens, the application aborts and returns an error message 4005.

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The Use of Crypto Tokens for In-App or In-Game Purchases

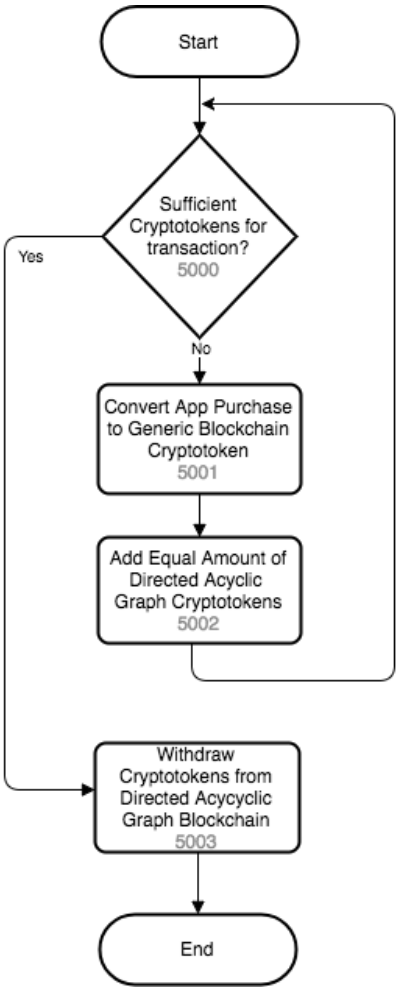


FIG. 5 is a flow diagram of how an application or game (App/Game) might use the Cryptotokens for in-app or in-game purchases. The App/Game first checks the wallet of the user or player to check for sufficient Cryptotokens for the transaction 5000. If there are enough Cryptotokens, a withdrawal occurs 5003. If there are not enough Cryptotokens, the user can add Cryptotokens by making an in-app or in-game purchase that deposits a certain percentage into the Generic Blockchain 5001 and then adds equal amounts to the Directed Acyclic Cryptotokens 5002 via a Cryptotoken wallet then rechecks the amount of Cryptotokens.

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Game in Progress for Proof of Concept



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Data Flow

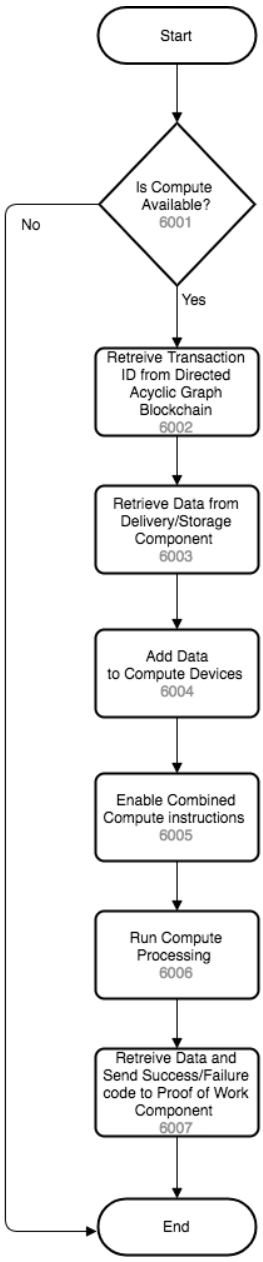


FIG. 6 is a flow diagram of the processing data flow. The first check is to make sure the devices have compute devices available for the processing of data. If not, the flow ends immediately. If it has compute devices available, the process retrieves the transaction ID from the Directed Acyclic Graph Blockchain system 6002. The process then retrieves the data from the Delivery/Storage Component 6003 and adds the data to the Compute Devices for processing 6004.

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Verification-Data Processing

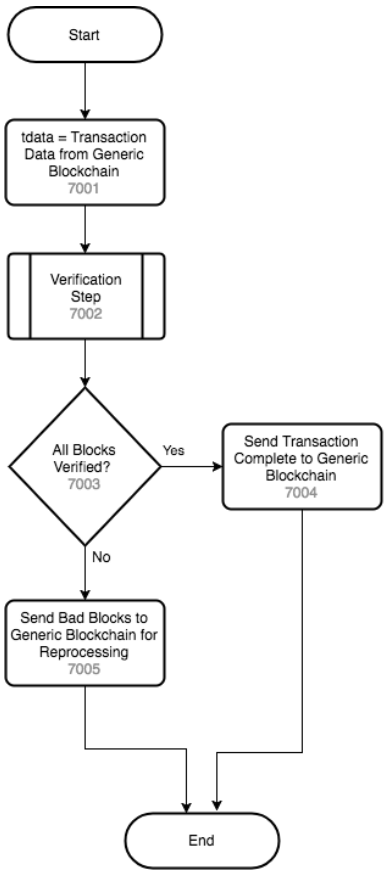


FIG 7 is a flow diagram of the verification step to check the processing of all data. This is also known as “Real Proof of Work.” The process first grabs the transaction data from the Generic Blockchain 7001 as data. It then sends data to the Verification Step 7002. If the verification step returns all blocks have been verified 7003, it sends a message to the Generic Blockchain that the transaction was completed successfully 7004. If all blocks have not been verified, the process then sends the bad blocks back to the Generic Blockchain for reprocessing 7005.

Verification-Blocks of a Transaction

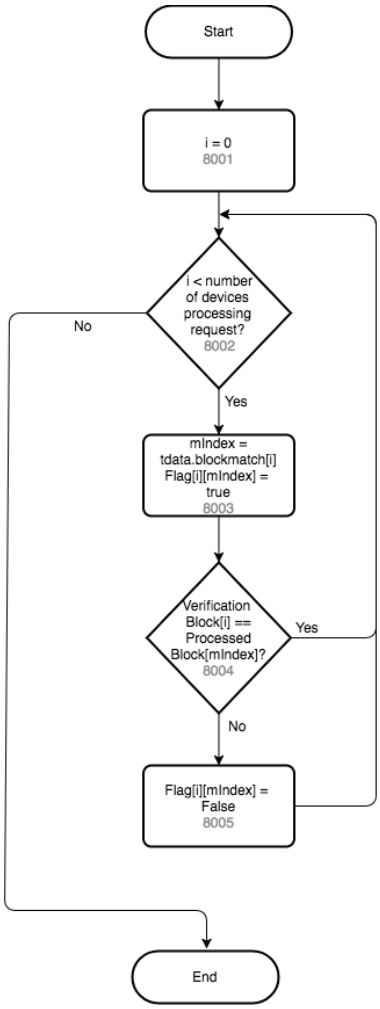


FIG 8 is a flow diagram of the verification step that verifies all blocks of a transaction request to process data. For each of the processing requests, a random block is checked from the verification device. A verification device is a random End User Device that receives a block from each of the other End User Devices and runs the processing of those blocks for verification. A counter ‘i’ is set to 0 as the first block index to check if counter ‘i’ is less than the number of devices involved in the processing request 8001. If yes, ‘i’ is used to index into a table of indexes that correspond to the random block that was chosen to verify if a device’s processed data and then that device is set to verified 8003. The checksum of the verification block is checked against the random block 8004. If the verification block checksum is equal to the system loops, check the next device that processed data. If the verification fails, a table index is marked as false for the failed device 8005. The process continues to verify all the devices' random verification blocks. If all the devices have been checked, the return value is an array of device verification pass or failure flags.

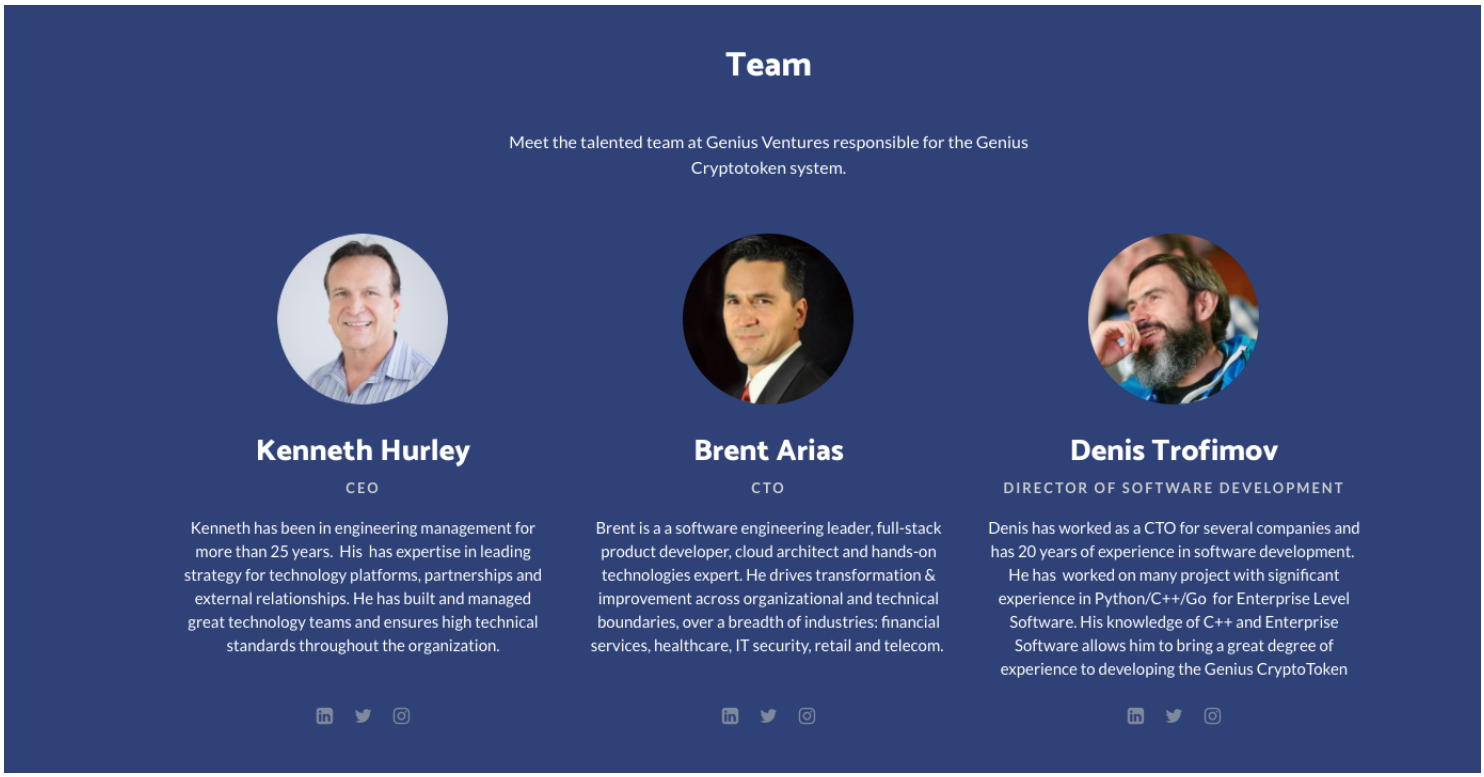
ITO

Initial Token Offering

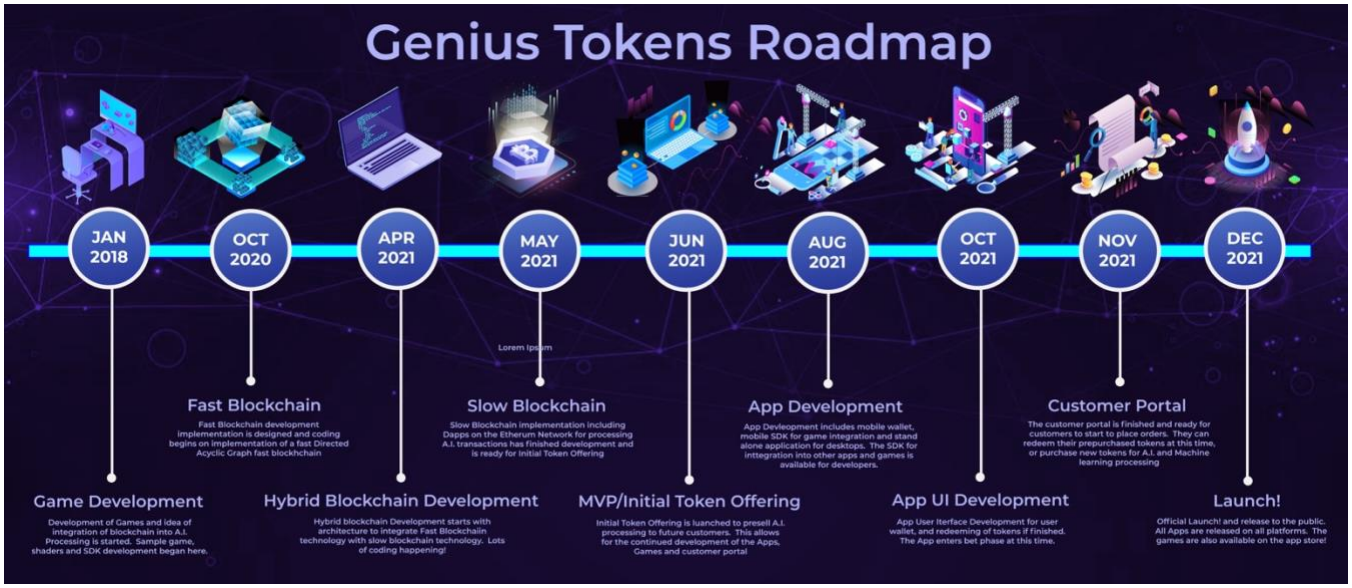
This is a non-regulatory means of raising funds for a new crypto token-based company. Start-up companies use an Initial Token Offering (ITO) instead of an Initial Coin Offering (ICO) to bypass a rigorous and regulated process required by U.S. government regulations.

At Genius Ventures, we aim to offer a percentage of the Genius tokens to our very first sponsors of this project in exchange for fiat currencies and other cryptocurrencies. The Genius Token ITO will be available after MVP. Please refer to our website https://www.geniusventure.io for more information.

Team



Roadmap



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References

Banafa, Ahmed (2017, Feb 27). IOT and Blockchain convergence. [Slides] Retrieved online from https://www.slideshare.net/professorbanafa/iot-and-blockchain-convergence

Factom - A Scalable Data Layer for the Blockchain http://fatcom.org/

[FogComputing] Cisco, Fog Computing and the Internet of Things: Extend

the Cloud to Where the Things Are, April 2015. https://www.cisco.com/c/dam/en_us/solutions/trends/iot/docs/computing-overview.pdf

Fedak, G., He, H., & Cappello, F. (2008, November). BitDew: a programmable environment for large scale data management and distribution. In IEEE International Conference for High Performance Computing, Networking, Storage and Analysis. SC 2008 (pp. 1-12).

Feldman,M.(2016,Nov 30).IDC'sLatestForecastfortheHPCMarket:"2016 isvLooking Good" [Article]. Retrieved from https://www.top500.org/news/idcslatest-forecast-for-the-hpc-market-2016-is-looking-good/

Gartner Says 8.4 Billion Connected «Things» Will Be in Use in 2017, Up 31 Percent From 2016, February 7, 2017, Egham, U.K. https://www.gartner.com/newsroom/id/3598917

He, H., Fedak, G., Tang, B., & Cappello, F. (2009, May). BLAST application with data-aware desktop grid middleware. In Proceedings of the 2009 9th IEEE/ACM International Symposium on Cluster Computing and the Grid (pp. 284-291). IEEE Computer Society.

[Cisco] D. Evans, "The internet of things: How the next evolution of the internet is changing everything," CISCO white paper, vol. 1, pp. 14, 2011.

International Data Corporation (2017, Sep 25) IDC Spending Guide Forecasts Worldwide Spending on Cognitive and Artificial Intelligence Systems to Reach $57.6 Billion in 2021 [Article] Retrieved from https://www.idc.com/getdoc.jsp?containerId=prUS43095417

Lebre, Adrien, Simonet Anthony, Orgerie , Anne-Cecile. Deploying Distributed Cloud Infrastructures: Who and at What Cost? Intercloud 2016, Apr 2016, Berlin, Germany. Proceedings of the fifth IEEE International Workshop on Cloud Computing Interclouds, Multiclouds, Federations, and Interoperability, 2016, http://www.cyclone-project.eu/intercloud2016.html

Mccann, C. (2018, April). State of the DApps: 5 Observations From Usage Data. Retrieved from: https://medium.com/@mccannatron/state-of-the-dapps-5-observations-from-usage-data-april-2018-a3e9da01bc22

Mircea Moca, Cristian Litan, Gheorghe Silaghi, Gilles Fedak (2016). Multi-criteria and satisfaction oriented scheduling for hybrid distributed computing infrastructures. Future Generation Computer Systems, 55, pp. 428-443.

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OpenFog Consortium (2017, Feb). OpenFog Reference Architecture for Fog Computing. [pdf] retrieved from http://www.openfogconsortium.org/wp-content/uploads/OpenFog_Reference_Architecture_2_09_17-FINAL-1.pdf

Sarmenta, L. F. (2002). Sabotage-tolerance mechanisms for volunteer computing systems. Future Generation Computer Systems, 18(4), 561-572.

Solomon, M. (2003). The ClassAd Language Reference Manual, Version 2.1. Computer Sciences Department, University of Wisconsin, Madison, WI, USA.

Stark, J. (February 2018). Making Sense of Ethereum's Layer 2 Scaling Solutions: State Channels, Plasma, and Truebit.